

GREEN COMPUTING & CLIMATE ACTION

Green Computing & Sustainable Technology

Carbon Footprints, Data Centres, AI-Driven Waste Management and the Individual Choices That Shape a Sustainable Digital Future



Sustainability



Data Centres



AI



Carbon Calc



Green Jobs

AGENDA

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What Is Green Computing?

Green computing (or green IT) is the practice of designing, manufacturing, using, and disposing of computers, servers, and associated devices in ways that minimize environmental impact — cutting energy use, reducing hazardous materials, and extending hardware life.



Reduce energy consumption across the device lifecycle



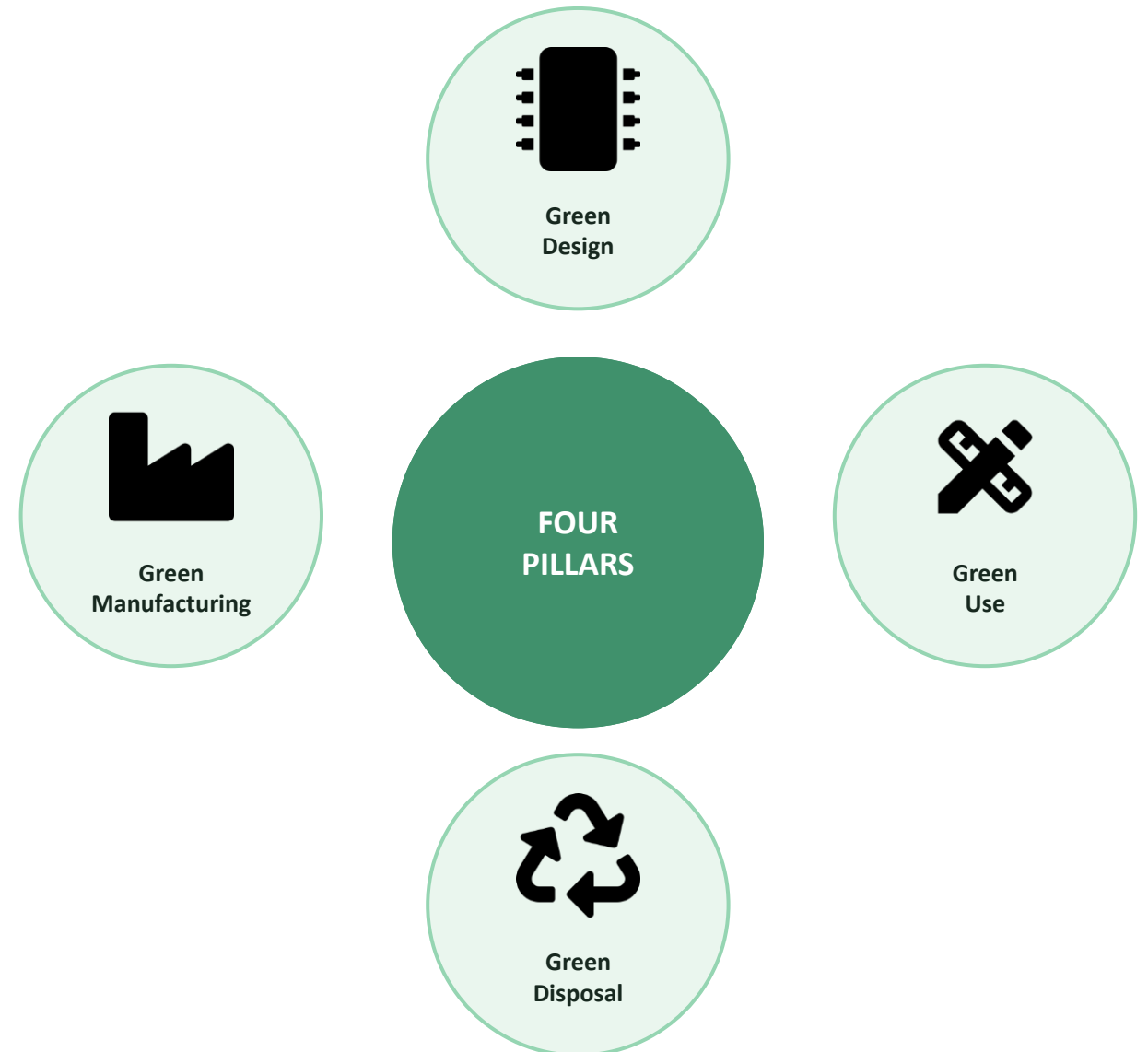
Minimize e-waste and hazardous material use



Extend hardware lifespan through reuse & upgrades



Favor renewable-powered infrastructure



Understanding Carbon Footprint

A carbon footprint is the total greenhouse gases (measured as CO₂ equivalent) generated directly and indirectly by an individual, organization, product, or event.



Scope 1 — Direct
Emissions from owned sources: vehicles, on-site fuel combustion

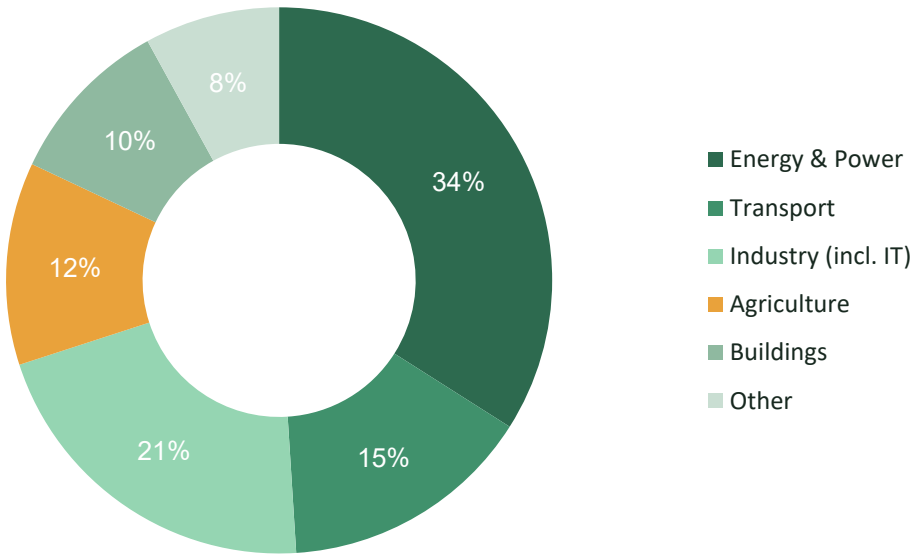


Scope 2 — Indirect Energy
Emissions from purchased electricity, heating & cooling



Scope 3 — Value Chain
Supply chain, business travel, product use & disposal

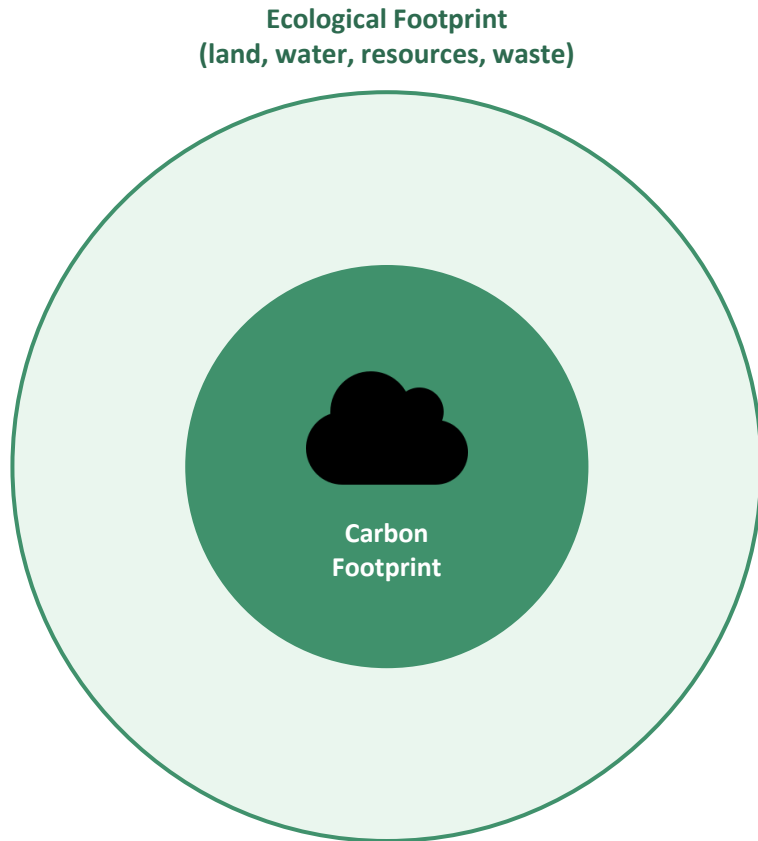
Illustrative Global GHG Emissions by Sector



Illustrative figures for classroom discussion, not a specific-year official dataset.

Ecological Footprint vs. Carbon Footprint

Ecological footprint measures the total land and water area required to produce the resources we consume and absorb the waste we generate. Carbon footprint is one component within this larger measure.



Carbon Footprint

Focuses solely on greenhouse gas emissions (CO₂e) from energy use, transport, and industry.



Ecological Footprint

Broader measure spanning land use, water consumption, food, forestry & waste absorption capacity.

Key takeaway: reducing your carbon footprint helps — but a truly sustainable lifestyle addresses the full ecological footprint.

Data Centres & Energy Demand

~1-1.5%

of global electricity use

40%

of data centre power spent on cooling

2x

projected demand growth by 2030

Why Data Centres Consume So Much

- Servers run 24/7 at partial load to guarantee uptime and redundancy
- Cooling systems offset heat generated by dense server racks
- Networking & storage scale continuously with cloud/AI demand

Greening the Data Centre



Liquid & free-air cooling



Renewable power purchase agreements



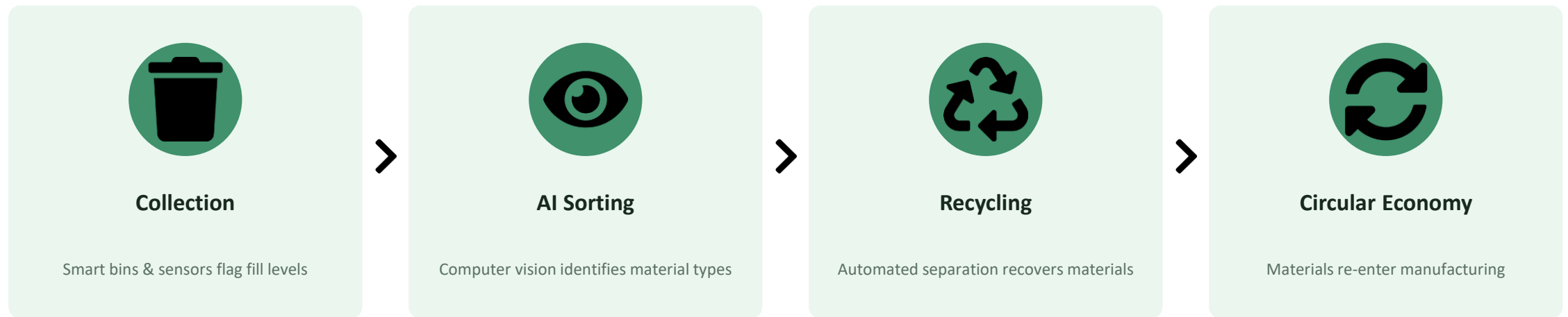
Server virtualization & consolidation



Siting near cold climates / clean grids

AI-Driven Waste Management

Artificial intelligence is transforming how waste is sorted, collected, and recycled — reducing landfill volume, recovering valuable materials, and closing the loop on electronic waste.



Where AI Adds the Most Value

- E-waste identification (batteries, circuit boards)
- Predictive collection routing cuts fuel use
- Contamination detection improves recycling yield

Sustainable Technologies



Renewable Energy

Solar, wind & hydro powering grids and facilities



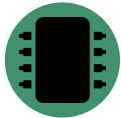
Green Cloud Computing

Shared, optimized infrastructure cuts per-user energy



E-Waste Recycling

Material recovery reduces mining & landfill demand



Energy-Efficient Hardware

Low-power chips & Energy Star certified devices



Smart Grids

Real-time balancing of supply, demand & storage



Biodegradable Materials

Compostable casings & sustainable packaging

Carbon Emission Calculators

Carbon calculators estimate greenhouse gas output from activities like travel, energy use, and digital consumption — turning abstract impact into actionable, comparable numbers.



Input Activity Data

Electricity bills, travel distance, device usage



Apply Emission Factors

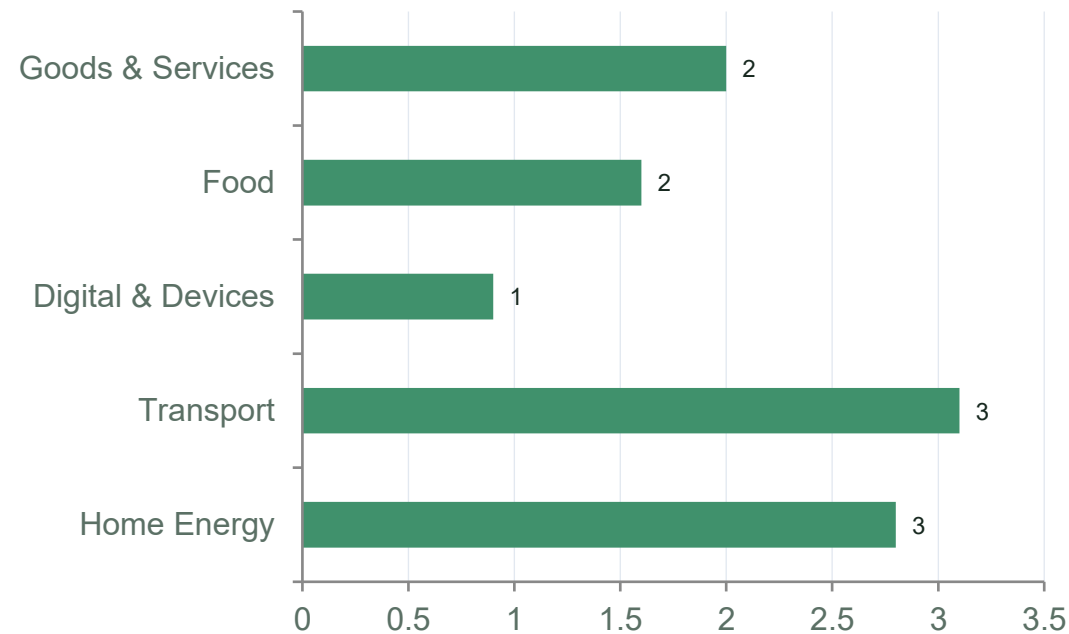
Standardized CO₂e values per unit of activity



Output: CO₂e Estimate

Total footprint, often with reduction tips

Illustrative Household Footprint Breakdown (tCO₂e / year)



Popular tools: EPA Carbon Footprint Calculator, WWF Footprint Calculator, CoolClimate Network.

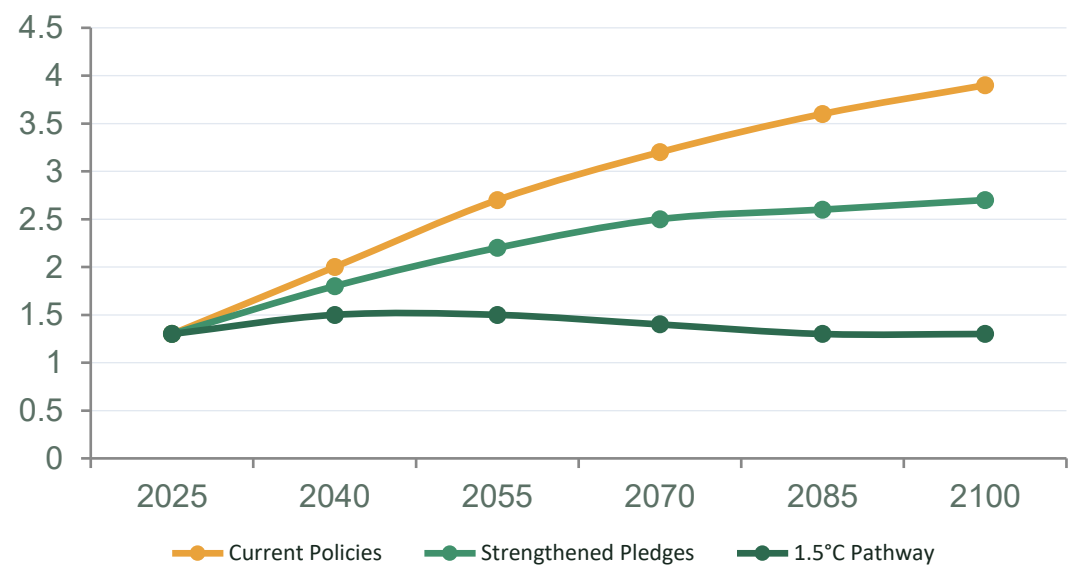
MIT En-ROADS Climate Simulator

En-ROADS is a free, interactive simulation developed by MIT Sloan and Climate Interactive that lets users test how energy, policy, and behavioral levers affect projected global temperature rise by 2100.

Policy Levers Users Can Adjust

-  Energy supply mix (coal, gas, renewables, nuclear)
-  Carbon pricing & subsidies
-  Deforestation & afforestation rates
-  Industrial energy efficiency
-  Transport electrification

Illustrative Temperature Trajectories to 2100



Classroom use: run live scenarios to show how combining multiple levers — not any single action — is needed to bend the curve toward 1.5–2°C.

Green Jobs & the Skill Council for Green Jobs (SCGJ)

SCGJ is a not-for-profit, industry-led skill development body (under India's Ministry of Skill Development & Entrepreneurship) that trains and certifies the workforce powering the renewable energy and sustainability sectors.



Solar Energy



Wind Energy



Green Buildings



Water Management



E-Mobility



Waste Management

Skilling

aligned with National Skill Qualification Framework

Industry-Led

governed by renewable & sustainability employers

Nationwide

training partners across Indian states

Why It Matters: The transition to sustainable technology creates entirely new career pathways — from solar installers to e-waste recyclers to sustainability analysts.

Individual Sustainability Practices



Unplug & Power Down

Fully shut down idle devices rather than leaving them on standby



Lower Screen Brightness

Reduces display energy draw significantly over time



Clean Up Cloud Storage

Delete unused files & emails to cut data-centre storage load



Choose Energy Star Devices

Buy certified energy-efficient hardware & appliances



Recycle E-Waste Properly

Use certified e-waste collection points, never landfill



Rethink Transport

Walk, cycle, carpool or use public transit when possible

Why This Matters: Key Statistics



~1-1.5%

of global electricity consumed by data centres



50M+ tonnes

of e-waste generated globally each year



<20%

of global e-waste is formally recycled



Rising Share

of data centre power now sourced from renewables

Figures are order-of-magnitude estimates commonly cited in sustainability literature (IEA, Global E-waste Monitor). Verify latest figures before formal citation.

A Roadmap Toward Greener Computing



Assess

Measure your footprint with
calculators & audits



Set Targets

Define reduction goals aligned with
1.5°C pathways



Adopt

Switch to efficient hardware, cloud
& renewable power



Advocate

Support green jobs, policy &
organizational change

Short-Term (0-6 mo)

Personal habits, calculator audit, device settings

Medium-Term (6-24 mo)

Efficient hardware upgrades, cloud optimization

Long-Term (2-5 yrs)

Renewable sourcing, policy advocacy, career shift



Technology and Sustainability Are Not a Trade-Off

From smarter data centres and AI-powered recycling to individual daily choices — every layer of the computing stack offers a lever for climate action. The tools exist. The jobs are growing. The choice is ours.

Thank You